

LOOM TRUST ENGINE V2

1. Executive Overview

This document defines Loom's enterprise-grade Trust Engine. It is a real-time behavioral intelligence system that continuously evaluates user risk and trust. It is internal-first, human-overridden, and designed to prevent harm before it occurs.

2. Core Philosophy

Trust is dynamic. Behavior outweighs identity. Systems must detect patterns, not isolated actions. No irreversible action is taken without human verification.

3. System Architecture

The system consists of 5 layers: Data Collection Layer, Signal Processing Layer, Scoring Engine, Risk Classification Engine, Moderation Interface Layer.

4. Scoring Model

Trust Score = weighted combination of Verification, Behavior, Interaction Quality, Reports, Reputation, Temporal Decay. Scores update in real-time per event.

5. Signals and Inputs

Signals include message frequency, consent violations, report clustering, block frequency, vouch credibility, interaction success rate, and anomaly patterns.

6. Risk Classification

Users are internally classified into states: Stable (Green), Monitor (Yellow), Critical (Red). These are NOT publicly exposed.

7. Real-time Processing

Event-driven architecture using queues (Kafka/Redis). Each user action triggers recalculation pipelines.

8. Exploitation Resistance

Weighted reports, anomaly detection, trust graph validation, behavioral fingerprinting, and cooldown systems prevent abuse.

9. Moderation Integration

Yellow users are queued for priority review. Red users trigger immediate escalation workflows including potential temporary restrictions.

10. Edge Case Handling

False positives mitigated via decay + human override. Coordinated attacks detected via graph anomaly. New users protected from harsh penalties.

11. UX Integration

Users see soft signals only: 'New', 'Building Trust', 'Highly Trusted'. No negative labels shown publicly.

12. Legal Safeguards

No automated punitive actions without review. Transparent appeal system. GDPR-compliant explainability layer.

13. Future Enhancements

Machine learning risk prediction, cross-user behavioral modeling, adaptive weighting, federated safety intelligence.

14. Conclusion

This system is Loom's core safety backbone. It must evolve continuously and be tested against adversarial behavior.